

Rational Inequalities

Date_____ Period____

Solve each inequality.

1) $\frac{x-7}{x-1} < 0$

2) $\frac{x+5}{x-4} \leq 0$

3) $\frac{x+32}{x+6} \leq 3$

4) $\frac{x+68}{x+8} \geq 5$

5) $\frac{(x+3)(x+5)}{x+2} \geq 0$

6) $\frac{x+6}{x^2-5x-24} \geq 0$

7) $-\frac{10}{x-5} \geq -\frac{11}{x-6}$

8) $-\frac{3}{x+7} \leq -\frac{4}{x+8}$

9) $-\frac{7}{x+5} \leq -\frac{8}{x+6}$

10) $\frac{(x+7)(x-3)}{(x-5)^2} > 0$

Critical thinking question:11) Write a rational inequality with the solution: $(-2, -1) \cup (1, \infty)$